

Below is an **advanced, full- scale project user requirements specification** for a **Zapier (low- code) capstone**. It's designed for a course setting and demonstrates **enterprise- grade orchestration** across **CRM, ERP, and SaaS apps**—including **actions via SaaS REST APIs**—and makes deliberate use of **Zapier Looping, Schedule, Filter, Formatter, and Paths**.

Outcome: A cohesive “Lead- to- Cash & Retain” automation program that moves data and work across **Salesforce, HubSpot, Dynamics 365, Zoho, Pipedrive, NetSuite, Shopify, Zendesk, Marketo, Mailchimp, Klaviyo, ActiveCampaign, Airtable, Asana, ClickUp, Monday.com, Trello, Wrike, Jira, Slack, Buffer, Hootsuite** (+ REST APIs).

Deliverable: A set of Zaps and governance artifacts meeting the requirements below, with **≥ 40 enumerated steps** (modularized into sub- zaps/paths).

1) Business Goal & Narrative

A B2B/B2C company wants to unify **lead acquisition → marketing nurture → sales routing → order & invoice → onboarding → support → renewal**, while keeping all systems synchronized and auditable. The program must reduce manual work, speed up response times, improve data quality, and provide leadership- grade daily KPIs.

2) Actors & Primary Systems

- **Marketing:** HubSpot, Marketo, ActiveCampaign, Mailchimp, Klaviyo, Buffer, Hootsuite
 - **Sales/CRM:** Salesforce, Dynamics 365, Zoho CRM, Pipedrive, HubSpot CRM
 - **Commerce/ERP:** Shopify, NetSuite
 - **Support & Success:** Zendesk, Freshworks, Jira
 - **Projects/Tasks:** Asana, ClickUp, Monday.com, Trello, Wrike, Jira
 - **Data/Comms:** Airtable, Google Sheets (optional), Slack
 - **Zapier Utilities:** **Schedule, Filter, Formatter, Paths, Looping, Webhooks by Zapier (Custom Request)**
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3) Data Model (Core Fields to Harmonize)

- **Person:** email, first_name, last_name, phone, country, consent (opt_in), UTM_* (source/medium/campaign), lead_score
- **Company/Account:** domain, name, industry, employee_count, ARR tier, region
- **Deal/Opportunity:** amount, currency, stage, owner, pipeline, expected_close_date
- **Order:** order_id, items[], subtotal, discounts, total, payment_status, fulfillment_status

- **Ticket:** priority, status, product, tags, SLA_due, csat
 - **Project Task:** title, description, due_date, assignee, priority, link_back
 - **Metadata:** source_system, timestamps, sync_status, error_reason
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4) Non- Functional Requirements

- **Reliability:** Idempotent upserts, retries, dead- letter queue (Airtable).
 - **Security:** OAuth where possible; API keys in Zapier connections; PII minimized in Slack.
 - **Observability:** Slack alerts on failures; daily KPI snapshot with diffs.
 - **Governance:** Naming conventions, versioning, staging → prod promotion.
 - **Rate Limits:** Looping + delays to respect vendor limits; pagination where needed.
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5) High- Level Architecture

A suite of Zaps grouped by domain, with standardized **ingress/egress** rules. **Paths** branch by region/product/tier; **Filters** gate quality; **Formatter** normalizes data; **Looping** handles arrays (line items, social posts); **Schedule** runs recurring jobs; **Webhooks by Zapier** hits **SaaS REST APIs** for actions not covered by native apps.

6) Detailed Feature Requirements & Step Inventory (≥ 40 steps)

The steps below are grouped into 8 cohesive automations. Each line is a **Zap step** (trigger or action). Cumulatively **42+ concrete steps** are specified.

A) Lead Intake, UTM Attribution & Enrichment (9 steps)

1. **Trigger – HubSpot:** New form submission.
2. **Formatter (Text/Utilities):** Normalize email case; split full name; parse UTM params.
3. **Filter:** Continue only if email present AND not from disposable domain.
4. **Salesforce (Find or Create Contact):** Upsert contact by email; attach UTM & source.
5. **Pipedrive (Create/Update Deal):** Create a “Qualification” deal if MQL criteria met.
6. **Klaviyo (Add/Update Profile):** Add to list with tags from UTM.
7. **Airtable (Create Record):** Write raw + transformed payload for audit.
8. **Slack (Message):** Post summary to #marketing- leads (PII redacted).
9. **Webhooks by Zapier (Custom Request):** **POST** to **Marketo REST API** /rest/v1/leads.json to ensure Marketo audience sync (include access token header).

Example Custom Request (Marketo):

1 POST https://<MARKETO_INSTANCE>.mktoREST.com/rest/v1/leads.json
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```

2  Headers:
3    Authorization: Bearer {{Marketo_Access_Token}}
4    Content-Type: application/json
5  Body (JSON):
6  {
7    "action": "createOrUpdate",
8    "lookupField": "email",
9    "input": [
10
11    {"email": "{{email}}", "firstName": "{{first_name}}", "lastName": "{{last_name}}", "utm_source": "{{utm_source}}", "utm_medium": "{{utm_medium}}", "utm_campaign": "{{utm_campaign}}"}
12  ]
13  }

```

B) Consent & Nurture Orchestration (Paths) (6 steps)

1. **Paths:** Branch by consent (opt- in true/false) and region (EU/US).
 2. **Mailchimp (Add to Audience):** EU + opt- in → add with GDPR tag & double- opt- in enabled.
 3. **ActiveCampaign (Add to Automation):** US + opt- in → enroll in 7- day welcome series.
 4. **Formatter (Date/Time):** Compute re- permission date (today + 365 days).
 5. **Airtable (Update Record):** Store consent regime & re- permission date.
 6. **Slack (Message with Approval Button):** If no consent → request manual review by compliance.
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C) Sales Routing Across CRMs (Looping, Filters) (8 steps)

1. **Zapier Schedule:** Every 15 minutes – poll for **new/updated leads** in **Zoho CRM** and **Dynamics 365** (use separate searches).
 2. **Looping – For Each Array:** Iterate over collected leads (merged array).
 3. **Filter:** Proceed only if ICP match (industry in [SaaS, FinServ] AND employee_count ≥ 50).
 4. **Salesforce (Create/Update Lead):** Harmonize into Salesforce as system of record.
 5. **HubSpot CRM (Create/Update Contact):** Maintain marketing alignment.
 6. **Paths (Territory):** Route by region (AMER → AE_A; EMEA → AE_B; APAC → AE_C).
 7. **Slack (DM):** Notify assigned AE with contact card + next best action.
 8. **Airtable (Create Record):** Log routing outcome with owner & timestamp.
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D) eCommerce → CRM/ERP Sync & Recovery (7 steps)

1. **Trigger – Shopify:** New paid order.
 2. **Formatter (Utilities):** Extract/flatten line items; compute total discounts; convert currency if needed.
 3. **Looping – Line Items:** Iterate to build ERP line array.
 4. **NetSuite (Create Sales Order):** Create SO with customer, items, taxes, memo.
 5. **Salesforce (Create/Update Opportunity):** Upsert Opp linked to Account, stage “Closed Won.”
 6. **Klaviyo (Profile Event):** Track “Purchased” with order_id & items.
 7. **Slack:** Post fulfillment handoff summary to #ops- fulfillment.
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E) Support Escalation with SLA (Paths, Filters) (6 steps)

1. **Trigger – Zendesk:** Ticket updated.
 2. **Filter:** Proceed if priority = urgent OR SLA_breached = true.
 3. **Jira (Create Issue):** Create bug or incident with custom fields (severity, product).
 4. **Freshworks (Create Ticket):** Mirror in Freshdesk/Freshservice for IT tracking.
 5. **Slack (Message):** Post to #support- warroom with **Block Kit** linkbacks.
 6. **Airtable (Create Record):** Append to escalation ledger for post- mortems.
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F) Order- to- Cash & Invoicing (REST API, Delay) (5 steps)

1. **Trigger – Salesforce:** Opportunity moved to “Closed Won.”
2. **NetSuite (Create Invoice):** Convert SO→Invoice or create Invoice directly (if configured).
3. **Webhooks by Zapier (Custom Request):** **POST** to **Stripe API** /v1/payment_links to generate a payment link (for manual invoices).
4. **Formatter (Numbers/Date):** Calculate due date + late fee thresholds; format currency.
5. **Slack:** Notify #finance with invoice + payment link and due dates.

Example Custom Request (Stripe):

```
1 POST https://api.stripe.com/v1/payment_links
2 Headers:
3   Authorization: Bearer {{stripe_secret_key}}
4 Body (form-encoded):
5   line_items[0][price]={{stripe_price_id}}
6   line_items[0][quantity]=1
7   metadata[crm_opportunity_id]={{sf_opportunity_id}}
```

G) Onboarding Project Spawner (Project PM Suite) (5 steps)

1. **Trigger – NetSuite:** Invoice paid (or Shopify fulfillment complete).
 2. **Paths (Customer Tier):** Enterprise → Asana project from template; SMB → Monday.com item + subitems; Lite → Trello board+cards.
 3. **Wrike (Create Tasks):** Always create Wrike tasks for onboarding checklist.
 4. **ClickUp (Create Task):** Create “Kickoff” + “Provisioning” tasks for IT.
 5. **Slack:** Announce new onboarding space and owners in #customer- launch.
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H) Social Amplification & Advocacy (Looping, Schedule) (5 steps)

1. **Trigger – Schedule:** Every weekday at 10:00 AM.
 2. **Airtable (Find Records):** Fetch approved content rows with status = ready and publish_date = today.
 3. **Looping – For Each Row:**
 4. **Buffer (Create Update)** with short caption; **Hootsuite (Create Post)** with long caption + link UTM.
 5. **Slack:** Post calendar to #social with queued links.
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I) Daily Executive KPI Digest (Formatter, Webhooks) (5 steps)

1. **Trigger – Schedule:** Daily at 07:45 AM local time.
2. **HubSpot, Salesforce, Shopify, Zendesk (Find/Search):** Pull yesterday’s metrics (MQLs, SQLs, revenue, CSAT).
3. **Formatter (Numbers):** Compute CAC est., ROAS, MQL→SQL rate, AOV, NPS delta.
4. **Airtable (Upsert):** Write metrics row; maintain 30- day rolling window.
5. **Slack:** Send KPI snapshot with trends; include **Webhooks by Zapier** call to **Looker Studio/BI API** (or internal metrics service) to refresh tiles if applicable.

Total steps enumerated: 56

7) REST API Usage Requirements

- **Use “Webhooks by Zapier → Custom Request”** whenever a native Zapier action is missing or limited.
- **Authentication:** Bearer token or API key headers; for OAuth APIs (e.g., Marketo, Slack, Shopify), supply tokens generated outside Zapier or via native app connections if available.
- **Error Handling:**

- **Retry** on 429/5xx with exponential backoff (Zapier auto- retries + a **Delay After Queue** pattern if needed).
 - On non- retryable errors, **write payload to Airtable “dead- letter”** with error_code, response_body, retry_after.
 - **Slack alert** on error with a masked payload ID.
 - **Pagination:** If the API returns lists (e.g., lead exports), use **Looping** over next cursors; keep a **cursor** in Airtable to resume.
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8) Governance & Quality

- **Naming:** CAP-<Domain>-<Purpose>-<Env> (e.g., CAP-Sales-Routing-Prod).
 - **Versioning:** Clone Zaps to -Staging for edits; peer review before promotion.
 - **Observability:**
 - Dedicated #automation- alerts channel for failures.
 - **Weekly roll- up** of dead- letter items and MTTR.
 - **PII Hygiene:** Redact emails/domains in Slack; store full PII only in CRMs/ERP.
 - **Rate- Limit Policies:** Maintain **per- app throughput caps** via **Delay** steps when Looping.
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9) Acceptance Criteria (Sampling)

1. **Lead Unity:** A new HubSpot lead becomes a **Salesforce Lead, Klaviyo profile, Airtable log, and Marketo lead** within **≤ 60 seconds** (p50), with **UTM fidelity** intact.
 2. **Consent Compliance:** EU opt- in always triggers **Mailchimp double opt- in**; non- consented records do **not** enter any marketing automation.
 3. **Routing SLA:** ICP leads from Zoho/Dynamics are **owned** in Salesforce and **notified** in Slack within **15 minutes** (scheduled batch).
 4. **Order Flow:** Shopify paid orders create **NetSuite SO** and **Salesforce Opp** with accurate line items and amounts.
 5. **SLA Escalation:** Zendesk urgent or SLA- breach ticks generate **Jira issues** and Slack war- room posts in **< 2 minutes**.
 6. **Cash Cycle:** Closed- Won Opp → **NetSuite invoice + Stripe payment link** created; finance receives Slack summary.
 7. **Onboarding:** Payment triggers **project creation** in one of Asana/Monday/Trello by tier, plus Wrike & ClickUp tasks.
 8. **KPI Digest:** Leadership receives a 07:45 AM Slack digest with yesterday’s metrics, including trend math.
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10) Field Mapping (Illustrative)

- **Email** → contact.email (Salesforce, HubSpot, Marketo, ActiveCampaign, Mailchimp, Klaviyo)
 - **Name** → split via **Formatter** → first_name, last_name
 - **UTM_fields*** → CRM custom fields + marketing tags
 - **Company domain** → Account/Company website & dedupe key
 - **Deal amount** → normalized currency (Formatter) before CRM/ERP write
 - **Shopify line items** → Loop → NetSuite item, quantity, rate
 - **Zendesk ticket priority** → Map to Jira priority via lookup table (Airtable)
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11) Error Handling & Recovery

- **Per- step “Continue on Failure”** only on non- critical fan- out (e.g., secondary marketing sync), writing to Airtable queue for later replays.
 - **Replay Zap:** A dedicated Zap retrieves dead- letter rows from Airtable and **replays** failed actions with backoff.
 - **Human- in- the- loop:** Slack interactive message (button) to **re- try now** or **snooze**.
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12) Environment & Secrets

- **Connections:** Separate **Staging** vs **Production** connections for Salesforce, NetSuite, Shopify, etc.
 - **API Tokens:** Rotated quarterly; scoped minimally (read/write principle of least privilege).
 - **IP Allowlisting:** If required by ERP/BI, work through VPN or allowlisted egress IPs.
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13) Course Project Deliverables

1. **Zap Inventory** with links, purpose, owners, and connection references.
 2. **Data Dictionary** (fields + systems of truth).
 3. **Swimlane Diagram** (textual or visual) explaining the end- to- end flow.
 4. **Demo Dataset** in Airtable to test each branch/path.
 5. **Test Plan** (happy paths + failure injections: 429, 5xx, validation errors).
 6. **Ops Runbook** (how to replay, rotate keys, extend mappings, add regions).
 7. **KPI Snapshot Examples** posted to Slack for 3 consecutive days.
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